

# ScriptLens Core Engines — Technical Review & Improvement Report

|                |                                                                                                                              |
|----------------|------------------------------------------------------------------------------------------------------------------------------|
| Scope          | scene_dependency.py, plot_contradiction.py, scriptlens_analyser.py, pdf_screenplay_loader.py                                 |
| Date           | June 13, 2026                                                                                                                |
| Audience       | Engineering (you) — what the engines do today, where they are weak, and what to fix next                                     |
| Status of code | Tier 1 + Tier 2 contradiction and full dependency engine implemented; recent Caveat-1 and Caveat-2 (partial) fixes committed |

## 1. Executive summary

ScriptLens Core is a **deterministic, rule + NLP screenplay analysis backend**. It has two engines:

1. **Scene Dependency Engine** — builds a directed graph of how scenes rely on each other (shared characters, props, locations) and answers "what breaks if I cut this scene?"
2. **Plot Contradiction Engine** — extracts structured facts and flags continuity conflicts (dead-then-alive, timeline, role/profession, object possession).

Both are **working and passing their current test suites** (contradiction precision/recall/F1 = 1.00 on the bundled fixtures). However, both rely heavily on **regex patterns + spaCy en\_core\_web\_sm**, and that creates a consistent class of weaknesses: **narrow pattern coverage (recall gaps)** and **occasional over-capture (precision gaps)**. The single highest-leverage area to invest in next is a **labeled evaluation corpus**, because almost every remaining improvement (recurrence promotion, grammatical-role detection, Tier-2 thresholds, Tier-3 LLM) needs real precision/recall numbers to tune safely.

## 2. What the engines do today

### 2.1 Pipeline (orchestration)

`scriptlens_analyser.analyze_screenplay()` runs:

```
screenplay_text
→ SceneDependencyEngine.parse_fountain_text()    # scenes: characters/objects/locations
→ SceneDependencyEngine.build_graph()             # directed dependency graph
→ ContradictionEngine.run_analysis()              # facts → Tier 1 → Tier 2 → dedup
→ assemble report (summary, high-risk scenes, contradictions, health score)
```

Input can be Fountain/plain text or PDF (`pdf_screenplay_loader.py` → text via `PyMuPDF/fitz`).

### 2.2 Scene Dependency Engine — current behavior

**Parsing (`parse_fountain_text`)** - Splits the script on scene headings: `INT.`, `EXT.`, `INT/EXT.`, `I/E..` - Per scene it extracts:

|  |
|--|
|  |
|--|

|                        |                                                                                                                                                                                                                                                                                                  |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Characters</b>      | (a) ALL-CAPS dialogue cues; (b) caps spans in action that match a known cue; (c) multi-word caps spans starting with a <b>title</b> (DETECTIVE, AGENT, ...); (d) <b>structural names</b> from fact phrasing ("KESSLER is a surgeon", "COLE is dead") — <i>added in the recent Caveat-2 fix</i> . |
| <b>Objects (props)</b> | (a) ALL-CAPS spans in action (screenwriting convention); (b) <b>objects of ownership verbs</b> "picks up / has / gives ... to" even when lowercase (e.g. "the blue ledger") — <i>Caveat-1 fix</i> ; (c) later lowercase mentions matched back to a known prop by exact or suffix match.          |
| <b>Locations</b>       | Primary place from the heading (text before _), uppercased.                                                                                                                                                                                                                                      |

**Graph (build\_graph)** - Directed graph; a node per scene. - "**First-seen**" model: when an entity first appears it is recorded; each later scene that reuses it gets an edge **from the first scene**. - Edge weights: character 1.0, object 0.7, location 0.4 (fact 0.5 defined but **unused**). Multiple signals between the same pair sum their weights.

**Queries** - get\_scene\_dependencies (upstream / ancestors), get\_delete\_impact (downstream / descendants), get\_orphan\_scenes (in-degree 0, excluding scene\_001), export\_graph\_summary, plus high-risk ranking in the analyser.

## 2.3 Plot Contradiction Engine — current behavior

**Fact extraction (extract\_facts)** — from **action lines only** (timeline also reads dialogue):

|                  |                                                                     |                                                      |
|------------------|---------------------------------------------------------------------|------------------------------------------------------|
| character_status | "X is/was dead/killed", "X died"                                    | DEAD_IDIOM_FOLLOWERS ("dead tired", "dead ahead"...) |
| character_trait  | "X is a ", "X works as ", one bespoke "the city's best ..." pattern | GENERIC_TRAIT_TERMS ("good man", "kid"...)           |
| timeline         | "Today/It is ", "Yesterday was ", "N days later"                    | TIME_OF_DAY_FOLLOWERS, flashback markers             |
| object_ownership | "picks up / has / gives ... to"                                     | dialogue excluded                                    |
| location         | heading + "the was/had been "                                       | —                                                    |

### Detection tiers

|               |                                                                                                                                        |                        |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>Tier 1</b> | Deterministic rules: character_alive_status, timeline_consistency, character_trait_conflict, object_ownership (with handoff detection) | Implemented            |
| <b>Tier 2</b> | spaCy similarity between same-entity/same-type facts below a threshold; opposing-state term boost; skips Tier-1-covered pairs          | Implemented            |
| <b>Tier 3</b> | Claude Haiku for ambiguous cases                                                                                                       | <b>Not implemented</b> |

**Health score:**  $100 - (\text{contradictions} \times 8) - (\text{orphans} \times 3)$ , floored at 0.

## 3. Recent improvements (this work)

|                                                 |                                                                                                                                                                                                                                                                                                                                            |         |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| <b>Caveat 1</b> — lowercase props missed        | Objects of ownership/handling verbs are now tracked even when never capitalized (e.g. "the blue ledger"), with suffix-based canonicalization of later mentions.                                                                                                                                                                            | 485db26 |
| <b>Caveat 2 (partial)</b> — cue-less characters | Title lexicon + title-case NER re-check (already present) <b>plus</b> new structural detection: the subject of "is dead / died / works as / is a <code>&lt;role&gt;</code> " is treated as a character, so a cue-less ALL-CAPS name is no longer misclassified as a prop. Pronoun / inanimate-death / generic-role guards keep it precise. | 4d1c8a4 |

## 4. Caveats & limitations (where the engines are weak)

### 4.1 Scene Dependency Engine

|    |                                                                                                                                                                                                                                                            |                                                                                  |         |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|---------|
| D1 | <b>Tiny ownership-verb list</b> (only <code>picks up</code> , <code>has</code> , <code>gives</code> ... <code>to</code> ). Handling verbs from the design (grabs, holds, hands, pockets, carries, takes, hides, steals, drops, sets down) are not covered. | Lowercase props introduced via other verbs are still missed.                     | Medium  |
| D2 | <b>NER-detected characters in action are excluded from props but never added as characters.</b> A cue-less, title-less name that <i>only</i> NER catches creates <b>no character edges</b> — it is silently dropped.                                       | Recall gap in the dependency graph.                                              | Medium  |
| D3 | <b>ALL-CAPS is exactly where spaCy NER is weakest</b> (root of Caveat 2). Signal 3 (grammatical-role / agentive-subject detection) is <b>not yet implemented</b> .                                                                                         | Invented names with no title and no fact phrasing are missed or mis-typed.       | Medium  |
| D4 | <b>First-seen-only edges.</b> Reuse links back to the <i>first</i> scene, not each prior scene. Deleting an intermediate scene won't show downstream breakage that conceptually flows through it.                                                          | <code>get_delete_impact</code> can under-state ripple effects on chained setups. | Medium  |
| D5 | <b>Suffix prop-matching can over-merge.</b> "THE KEY" could resolve to "SILVER KEY" <i>or</i> "MASTER KEY" (first match wins).                                                                                                                             | Wrong prop edges when two props share a head noun.                               | Low–Med |
| D6 | <b>fact and causal dependency edges are unimplemented</b> ( <code>fact</code> weight exists but is never produced; causal/dialogue edges from the spec are absent).                                                                                        | Story dependencies carried by dialogue ("after what you did") are invisible.     | Medium  |
| D7 | <b>Performance: up to 2 spaCy passes per scene</b> ( <code>nlp(action_text)</code> + a second pass on title-cased text), and the model is loaded <b>separately in each engine</b> .                                                                        | ~2× NLP cost; risk against the <400ms/<800ms SLA on 120-scene scripts.           | Medium  |
| D8 | <b>Locations are single-level</b> (primary place only; time-of-day and sub-locations dropped).                                                                                                                                                             | "INT. HOUSE - KITCHEN" and "INT. HOUSE - BEDROOM" collapse to one location.      | Low     |

### 4.2 Plot Contradiction Engine

|    |                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                |         |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|---------|
| C1 | <b>Entity over-capture / no pronoun guard in fact extraction.</b> The trait/status entity regex is lazy and allows periods, so on multi-sentence lines it can grab the wrong span, and there is no pronoun stoplist. <b>Observed real bug:</b> a run flagged a <code>character_trait_conflict</code> with entity "THERE" and garbage trait values ("long moment as Marty looks..." vs "similar rig on the real DeLorean"). | False-positive contradictions; erodes trust.   | High    |
| C2 | <b>Tier 2 similarity uses <code>en_core_web_sm</code>, which has no real word vectors.</b> <code>.similarity()</code> falls back to context/tagger vectors and is unreliable.                                                                                                                                                                                                                                              | Tier-2 semantic results are noisy/low-quality. | High    |
| C3 | <b>Narrow status/trait coverage.</b> Death = only "dead/killed/died" (misses "murdered", "passed away", "gone", "we lost him"). Trait = "is a / works as" + one bespoke pattern (misses appositives "X, a doctor,", "became a lawyer").                                                                                                                                                                                    | Missed real contradictions (false negatives).  | Medium  |
| C4 | <b>No coreference resolution.</b> Facts about "he/she/they" are lost; only explicitly named subjects are captured.                                                                                                                                                                                                                                                                                                         | Large recall gap on natural prose.             | Medium  |
| C5 | <b>Timeline is weekday-only</b> (+ "N days later"). No calendar dates, "next morning", "that night", or relative ordering.                                                                                                                                                                                                                                                                                                 | Most real timeline conflicts go undetected.    | Medium  |
| C6 | <b><code>character_alive_status</code> appearance test is a substring match.</b> <code>_character_appears_in_scene</code> regex-matches the name in any line, which can match partial/incidental mentions and ignores legitimate flashback resurrection.                                                                                                                                                                   | Occasional false positives.                    | Low–Med |
| C7 | <b>Trait conflict = "no overlapping words".</b> Two non-conflicting roles with disjoint vocabulary ("detective" vs "negotiator") can be flagged.                                                                                                                                                                                                                                                                           | False positives on complementary roles.        | Medium  |
| C8 | <b>Health score weights are arbitrary</b> (8/contradiction, 3/orphan) and unvalidated against any rubric.                                                                                                                                                                                                                                                                                                                  | Score is a rough signal, not calibrated.       | Low     |

### 4.3 Cross-cutting / infrastructure

|    |                                                                                                                                                                                                                                             |                                                                     |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| X1 | <b>No automated pytest CI for the engines.</b> Tests are manual runner scripts ( <code>run_*_test.py</code> , <code>real_screenplay_test.py</code> ) with embedded screenplays; only <code>tests/test_pdf_loader.py</code> is pytest-style. | Regressions can slip in silently.                                   |
| X2 | <b>Evaluation corpus is essentially empty</b> (only <code>demo_real</code> ). The harness exists ( <code>scripts/run_corpus_batch.py</code> + ground-truth YAML template) but has no labeled data.                                          | Can't measure precision/recall, so risky heuristics can't be tuned. |
| X3 | <b>spaCy model loaded twice</b> (once per engine).                                                                                                                                                                                          | Extra memory + startup time.                                        |

## 5. Where to work next — prioritized roadmap

Ordering principle: **fix precision bugs first** (they directly hurt user trust), then **safe recall wins**, then **anything that needs tuning** (gate those behind the corpus).

## P0 — Precision fixes (do now, low risk, high trust impact)

1. **Fix contradiction entity over-capture (C1)**. Reuse the same guard approach already added to the dependency engine: extract the trailing proper-name span, and reject pronouns/indefinites (`NON_CHARACTER_WORDS`). This kills the "THERE is described as..." class of false positives.
2. **Add a pronoun/non-name guard to `character_trait` and `character_status` extraction (C1/C6)**.
3. **Constrain the trait entity regex** so it cannot span across sentence boundaries (don't allow . inside the captured entity, or anchor to line start).

## P1 — Safe recall wins (low/medium risk)

1. **Expand the interaction-verb list (D1)** — add grabs, holds, hands, pockets, carries, takes, hides, steals, drops, sets down. Or switch to spaCy dependency parse: direct objects of a handling-verb set. Share the list between both engines (already shared as `OBJECT_OWNERSHIP_PATTERNS`).
2. **Add death/status synonyms (C3)** — "murdered", "passed away", "killed off", "his body".
3. **Promote NER-detected action characters to scene characters (D2)** — close the dependency recall gap, gated by the existing person filters.

## P2 — Higher-value but needs the corpus first (tune against real numbers)

1. **Signal 3 — grammatical-role character detection (D3)**. Classify a caps span as a character when it is the `nsubj` of an agentive verb (vs `dobj/pobj` for props). This is the real fix for invented names NER misses, **but** it has known false-positive risk (e.g. "DELOREAN races") and needs threshold/lexicon tuning.
2. **Recurrence promotion for weak props** — a recurring lowercase noun chunk across 2+ scenes becomes a low-weight prop. Reintroduces a noise channel; only ship with corpus measurement.
3. **Upgrade Tier-2 to `en_core_web_md` (C2)** — real word vectors make semantic similarity meaningful; re-tune `TIER2_SIMILARITY_THRESHOLD`.
4. **Coreference resolution (C4)** — pronoun → entity, to recover dropped facts.

## P3 — Infrastructure & product

1. **Build the labeled corpus (X2)** — this unblocks P2. Use the existing `tests/corpus/ground_truth/_template.yaml` + `run_corpus_batch.py --compare-ground-truth`. Target ~10–20 annotated scripts.
2. **Add a pytest suite + CI (X1)** — convert the manual runners into assertions so precision/recall are checked on every change.
3. **Single shared spaCy load (D7/X3)** — load `en_core_web_sm` once and inject into both engines; reduce to one NLP pass per scene where possible.
4. **Implement fact/causal dependency edges (D6)** — the `fact` weight is already reserved.
5. **Tier 3 (Haiku) for ambiguous Tier-2 candidates** — only after the corpus shows where Tier 1+2 fall short.

## Suggested sequence

P0 (precision) → P1 (safe recall) → build corpus (P3.11) → P2 (tuned heuristics) → remain

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## 6. Evaluation infrastructure (already in place)

You don't have to build the harness from scratch — it exists and just needs data:

- `scripts/run_corpus_batch.py` — runs every script in `tests/corpus/input/`, writes per-script `*_report.txt`, `*.json`, and a `manifest.csv`. With `--compare-ground-truth` it diffs against YAML ground truth and reports matched / missed (false negatives) / extra (false positives) + simulate-delete checks.
- `tests/corpus/ground_truth/_template.yaml` — the annotation format (expected + planted contradictions, expected delete-impact).
- `scripts/benchmark_engines.py` + `docs/PERFORMANCE_TESTING_GUIDE.md` — latency/throughput measurement (relevant to D7 SLA concerns).

**Action:** annotate ~10–20 scripts to turn "I think this is a false positive" into a measured precision number. This single step de-risks every P2 item.

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## 7. File map (core)

```
scriptlensCore/
├─ scene_dependency.py      # parsing + dependency graph + queries
├─ plot_contradiction.py    # fact store + Tier 1 / Tier 2 rules
├─ scriptlens_analyser.py   # orchestration + health score + report
├─ pdf_screenplay_loader.py # PDF → Fountain-ish text (PyMuPDF)
├─ run_scriptlens.py        # CLI entry point
├─ run_dependency_test.py   # manual runner (sample script)
├─ run_contradiction_test.py # manual runner (accuracy fixture)
├─ real_screenplay_test.py  # end-to-end check with planted contradictions
├─ scripts/run_corpus_batch.py # batch eval + ground-truth diff
├─ tests/corpus/            # evaluation corpus (currently almost empty)
└─ docs/ARCHITECTURE.md     # full product/architecture spec
```

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## 8. One-paragraph takeaway

The engines are **functionally solid and well-guarded for their bundled fixtures**, but they are **pattern-bound**: recall is limited by the breadth of the regexes/verb lists, and precision is occasionally undermined by entity over-capture (the "THERE" trait bug) and by Tier-2 running on a vectorless spaCy model. **Fix the precision bugs (P0) immediately**, take the **safe recall wins (P1)**, then **invest in a labeled corpus** so the genuinely valuable but risky improvements (grammatical-role character detection, recurrence promotion, better semantic similarity, Tier 3) can be tuned with real numbers instead of guesswork.

*End of report.*